

Security Linter Report

Filename	case9_complex_propagation.py
Timestamp	2026-05-22 17:49:34 UTC
Lines of Code (LOC)	27
Verdict	VULNERABLE

ID	Severity	Sink	Source Variable	Source Type	Line
CVE-SL-002	WARNING	cursor.execute	user_input	STDIN	37
CVE-SL-003	WARNING	cursor.execute	query	STDIN	38

CRITICAL	0
WARNING	2
INFO	0
TOTAL	2

CVE-SL-002 — cursor.execute**WARNING**

CWE	CWE-89
Sink	cursor.execute
Sink Line	37
Sink Column	0
Sink Expression	cursor.execute(final_query)
Argument	final_query
Source Variable	user_input
Source Type	STDIN
Propagation Mechanism	direct assignment from stdin

Taint Path Trace:

Step	Role	Line	Expression
1	SOURCE	6	user_input = input("Enter search: ")
2	ASSIGN	16	final_query = "SELECT * FROM posts WHERE " + search_terms
3	SINK	37	cursor.execute(final_query)

Remediation:

Use a parameterized query and validate the input type. If the value must be an integer, cast it with int():

```
user_id = int(input("ID: "))
query = "SELECT * FROM users WHERE id = ?"
cursor.execute(query, (user_id,))
```

CVE-SL-003 — cursor.execute**WARNING**

CWE	CWE-89
Sink	cursor.execute
Sink Line	38
Sink Column	0
Sink Expression	cursor.execute(query)
Argument	query
Source Variable	query
Source Type	STDIN

Propagation Mechanism	direct assignment from stdin
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Taint Path Trace:

Step	Role	Line	Expression
1	SOURCE	23	query = "INSERT INTO profiles (username, email, bio) VALUES ...
2	SINK	38	cursor.execute(query)

Remediation:

Use a parameterized query and validate the input type. If the value must be an integer, cast it with int():

```
user_id = int(input("ID: "))
query = "SELECT * FROM users WHERE id = ?"
cursor.execute(query, (user_id,))
```

No safe sink paths confirmed.

No sanitization functions detected.

CFG — Total Nodes	23
CFG — Condition/Loop Nodes	1
CFG — Assignment Nodes	14
CFG — Call Nodes	5
CFG — Merge Nodes	1
██	██████████
DFG — Total Nodes	68
DFG — Total Edges	67
DFG — Source Nodes	5
DFG — Sink Nodes	2
DFG — Operator Nodes	7
DFG — Constant Nodes	18
DFG — Variable Nodes	26
Taint Propagation Hops	24
██	██████████
Analysis Time (ms)	49.2

Vulnerabilities	0 CRITICAL 2 WARNING 0 INFO
Safe Sinks	0
Analysis Time	49.2 ms
Engine Version	1.0.0
Final Verdict	VULNERABLE